

Reflective Piece: Numerical Analysis

The module did not teach me to trust data more. It taught me to trust my own interpretations less until they had been tested through statistical scrutiny.

That realisation built up slowly across twelve increasingly demanding weeks. After more than twenty years in clinical operations and project management, AI governance became my role only recently, and my last real contact with quantitative methods dated back to my economics studies at HEC Lausanne. Coming back to formal statistics was disorienting in a way I had not expected. A lot of what I considered evidence-based reasoning in my professional life had never been checked statistically. I simply assumed it had. Gibbs (1988) writes that experience alone is not enough, and that without reflection, its learning potential is lost; this module kept proving him right.

The Scope of the Learning

The module moved from descriptive statistics and plots in R to hypothesis testing, regression and Bayesian reasoning. Every unit added a technique and a harder question about what the evidence could be asked to show. The artefacts in Appendix B follow this progression against the module learning outcomes, from cross-tabulations and chi-square through correlation and simple regression to Bayesian conditional probability.

Technical Development and Growing Discomfort

R gave me the most persistent difficulties. I had never programmed before, and the exactness it demands (one wrong variable name and nothing runs) caused real frustration early on. What I slowly understood is that R forces the reasoning into the open. Every assumption has to be written down, which also means it can be questioned. For the Unit 9 cross-tabulations, the brief specified grouping states into regions and banding cases into levels; implementing them in R still left the cut-offs to me, and the table shifted with every boundary I tried. The same applied to the ggplot2 scatter plots on the blood pressure data, where scale and annotation shape what a reader takes away. Unit 8 introduced the Wilcoxon and Kruskal-Wallis tests; understanding why they fit when normality cannot be assumed felt less like a workaround than learning to pick the correct tool over the comfortable one.

Peer Feedback and Revised Thinking

The formative discussion around Brown (1994) tested these habits in front of others. I had redesigned his Table 2 as a diverging stacked bar chart ranking the reasons discouraging GPs from intrapartum care, and argued that the data pointed towards the willing minority already involved. My peers were not convinced. One colleague, citing Kankaraš and Capecchi (2025), challenged where I had put the no-opinion answers. A neutral midpoint assumes balance, but those responses can just as easily mean uncertainty or reluctance to answer. Another saw a possible selection effect: with cross-sectional data, there is no way to tell whether home deliveries made those GPs willing, or whether they were willing to begin with. They were right on both counts, and seeing the data through their eyes showed me that my chart carried assumptions I never argued. The dot plot I raised in my initial post, separating certainty from direction, still looks the stronger design. Reviewing two peers' charts

put me on the other side, questioning their grouping and midpoint choices. Tutor feedback so far has come through the mid-module mathematics test (96%, a distinction); for the analysis work, peer critique and the weekly seminars have been my main external checks. Hatton and Smith (1995) rank critical reflection above dialogic reflection because it situates conclusions in a wider context; peer correction moved me in that direction.

Critical Transformation

Nothing changed my thinking more than the work on p-values and confidence intervals. I used to read a p-value as a verdict, significant or not, which is also how most governance documents I encounter treat it. Schober, Bossers and Schwarte (2018) took that reading apart for me: statistical significance and practical importance are different things, and p below 0.05 says nothing about how large an effect is. Confidence intervals turned out to be more useful for decisions, since a range of plausible values keeps uncertainty visible instead of collapsing it. When I reported the 95% interval for my regression slope (0.45 to 0.54, Appendix B), I learnt more from that range than from the p-value next to it.

I cannot separate this from my professional role. If a governance lead is unable to judge the method, they cannot judge whether the conclusion follows from it, and the review becomes ratification rather than oversight. My own governance questions used to stop at controls, ethical risk and regulatory compliance; I rarely asked how statistically sound the evidence underneath was. Informatica (2026) reports that 76% of organisations find their AI governance cannot keep pace with AI usage, a gap attributed to insufficient data literacy that I recognise.

The Bayesian unit pushed this furthest. Van de Schoot et al. (2021) describe Bayesian inference as the principled revision of prior beliefs in the light of new data, with uncertainty treated as part of reasoning rather than a flaw to remove. For AI governance, that framing feels more honest than threshold-based testing, since organisational decisions are rarely final and new evidence should move existing assessments.

New Practice and Professional Development

Rolfe, Freshwater and Jasper (2001) would ask what I do differently now. In governance contexts, I no longer let statistical claims pass unexamined. I ask which method was used, whether it suited the data, and what the result can support; the same questions now apply to AI model performance claims inside my organisation. Appendix A holds my updated skills matrix and action plan, benchmarked against my results in earlier modules, with an honest split between skills now solid and those still in progress.

My professional experience has not lost its value. Reading a situation quickly and framing the right strategic questions still comes naturally to me. What statistics added is a brake, a habit of separating a plausible explanation from a demonstrated one before acting on it. Looking back, the module has not made me trust data more. It has made me slower to trust my own interpretation of it, until the numbers have been given the chance to disagree.

References

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Appendix A: Professional Skills Matrix and Action Plan (PDP)

Self-assessment scale: Developing, Competent, Proficient. Baseline benchmarks:

Launch into Computing, 76 (October 2025); Understanding Artificial Intelligence, 87 (January 2026). Tutor feedback within this module to date: mid-module mathematics test, 96% (Distinction). Peer feedback given and received in the Unit 5 collaborative discussion is incorporated below. I have claimed a level change only where the module focused directly (numeracy and R skills); elsewhere, the improvement stayed within the same band, and the remaining gaps are what the upcoming modules and the dissertation should address.

Skill	Module start	Module end	Evidence	Action plan (PDP)
Time management	Competent	Competent (maintained)	Kept pace with the weekly R activities and formative tasks around a full-time role; the notes behind this reflection were written as the module went on, not rebuilt at the end.	For future modules, I will continue to set my own deadlines, protect weekly study slots from Unit 1 onwards, and scope the dissertation timeline early rather than under pressure.
Critical thinking & analysis	Competent	Competent (improved)	Reworked Brown's (1994) Table 2 and had to defend the encoding choices; revised my reading after a peer raised a selection effect; checked normality before choosing tests rather than defaulting to the familiar ones.	Carry the assumption-checking habit into future modules (evaluating and optimising models, what a model can and cannot claim) and into AI model validation evidence reviewed at work.
Communication & literacy	Proficient	Proficient (spoken still developing)	Three forum posts (initial, peer responses, summary) with full referencing; earlier written work marked at 87 (Understanding Artificial Intelligence).	Machine Learning is assessed partly through a project presentation; keep working on spoken delivery as a non-native speaker, and reuse the same plain-language approach

Skill	Module start	Module end	Evidence	Action plan (PDP)
				when presenting statistical evidence to leadership at work.
IT & digital	Developing	Competent	Working R scripts for Units 8 to 12 (readr, dplyr, haven, ggplot2), written and debugged with no prior programming background; started from a baseline of 76 in Launch into Computing.	Machine Learning will bring new programming demands (its seminars include live coding); transfer the habits learnt in R rather than starting from zero.
Numeracy	Developing	Competent	96% (Distinction) in the mid-module mathematics test; chi-square, correlation ($r = 0.847$), regression with confidence intervals, and Bayes theorem all computed and checked in R (Appendix B).	Regression and probability feed straight into Machine Learning (linear regression reappears in supervised learning, followed by model evaluation and optimisation); revisit both before Unit 1, and read model performance statistics at work with the same scrutiny.
Research	Competent	Competent (improved)	Went beyond the reading list to source Schober, Bossers and Schwarte (2018), van de Schoot et al. (2021) and Kankaraš and Capecchi (2025) for the forum work and this reflection.	Start a running literature log now: Research Methods and Professional Practice will expect a defined topic, and AI governance in clinical research is the likely dissertation direction.
Interpersonal	Proficient	Proficient	Took two corrections publicly in the forum and worked them into my summary post; gave detailed responses on two peers' charts in return, including questions they picked up.	Future modules are partly assessed on working in a virtual team; at work, run the planned Data & AI literacy sessions, where explaining data and AI to non-specialists will test the same skill.

Skill	Module start	Module end	Evidence	Action plan (PDP)
Problem-solving	Competent	Competent (improved)	Debugged data-type and missing-value errors in R without support; switched to Wilcoxon and Kruskal-Wallis when normality failed; implemented the Unit 9 groupings and saw how much the cut-offs changed the table.	Future modules will raise harder versions of the same problems (the challenges of different datasets, designing a machine learning system); professionally, apply the same troubleshooting mindset when AI tools under governance review behave inconsistently.
Ethical awareness	Proficient	Proficient	The Brown (1994) redesign showed me that category and chart choices shape conclusions; this connected directly to the data integrity principles (GCP, ALCOA+) I already work with.	Machine Learning and Research Methods both carry explicit ethics outcomes; professionally, build evidence-quality questions into the AI governance review criteria I maintain.

Appendix B: Artefacts Mapped to Module Learning Outcomes

Module learning outcomes. *LO1*: Demonstrate systematic understanding of key mathematical and statistical concepts underpinning Data Science and AI. *LO2*: apply mathematical and statistical methods to help in the decision-making process. *LO3*: Critically evaluate the use of statistical analysis and the numeric interpretation of results as aids in decision-making. *LO4*: Critically appraise and present results of a statistical analysis to a diverse audience.

Artefact 1 (Unit 8). Non-parametric testing on Health_Data: Wilcoxon rank-sum test comparing diastolic blood pressure between diabetic and non-diabetic participants ($W = 3804.5$, $p = 0.7999$) and Kruskal-Wallis test of systolic blood pressure across occupational groups ($\chi^2 = 0.79$, $df = 3$, $p = 0.8545$), with supporting boxplots; systolic blood pressure had previously been shown to violate normality. Evidence *LO2* and *LO3*: applying rank-based alternatives when parametric assumptions could not be met, and interpreting non-significant results without treating them as proof of no effect.

Is Median Diastolic BP the Same Among Diabetic and Non-Diabetic Participants?

We used: Mann-Whitney U test / Wilcoxon rank-sum test

Purpose

Compare median DBP between two independent groups
Non-parametric alternative to independent t-test
Useful when normality may not hold

Hypotheses

Null hypothesis (H0): Median DBP is the same between diabetic and non-diabetic participants.

Alternative hypothesis (HA): The location shift is not equal to 0.

Meaning: One group tends to have generally higher or lower DBP values than the other.

What the p-value means

$$p = 0.7999 > 0.05$$

This means:

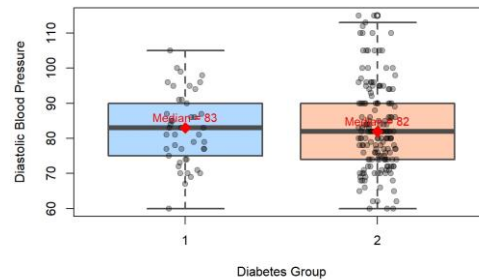
the result is not statistically significant
the null hypothesis is not rejected

If one group truly tended to have higher DBP values overall, you would usually see:
noticeably different medians
less overlap between boxplots
more shifted distributions
and usually: $p < 0.05$

But graph shows:

almost identical medians
strong overlap
very similar spread
So the non-significant result makes sense visually.

DBP by Diabetes Group



wilcoxon rank sum test with continuity correction
data: dbp by diabetes
W = 3804.5, p-value = 0.7999
alternative hypothesis: true location shift is not equal to 0

Is Systolic BP Different Across Occupational Groups?

This follows the Unit 8 Kruskal-Wallis approach because: more than two groups exist
SBP was previously shown to violate normality

Purpose

Compare SBP across several occupation groups
Non-parametric alternative to ANOVA
Uses ranks instead of raw means

- Medians are very similar
 - The boxplots overlap heavily.
- This means:**
- the SBP distributions are very similar
 - no occupation group clearly stands apart
 - variability within groups is larger than differences between groups

Some participants have very high SBP values.
outliers appear across several groups BUT they do not create a clear occupational pattern

Chi-squared statistic

This measures how different the ranked SBP distributions are across groups.
A very small value suggests: little difference exists between groups

p-value $p = 0.8545 > 0.05$ the result is not statistically significant + the null hypothesis is not rejected

Hypotheses

Null hypothesis (H0): SBP has the same central tendency across occupational groups

Alternative hypothesis (HA): At least one occupational group has a different SBP distribution

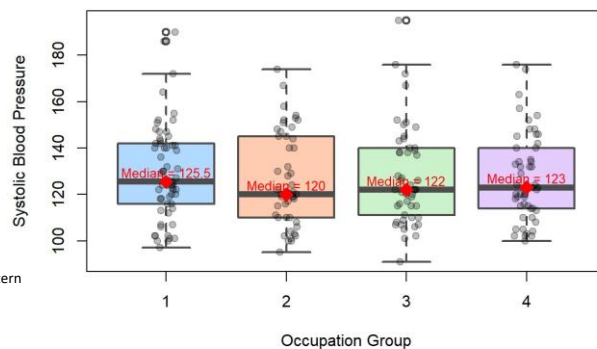
Final conclusion

The Kruskal-Wallis test showed no statistically significant difference in systolic blood pressure across occupational groups.

The graph visually supports this because:

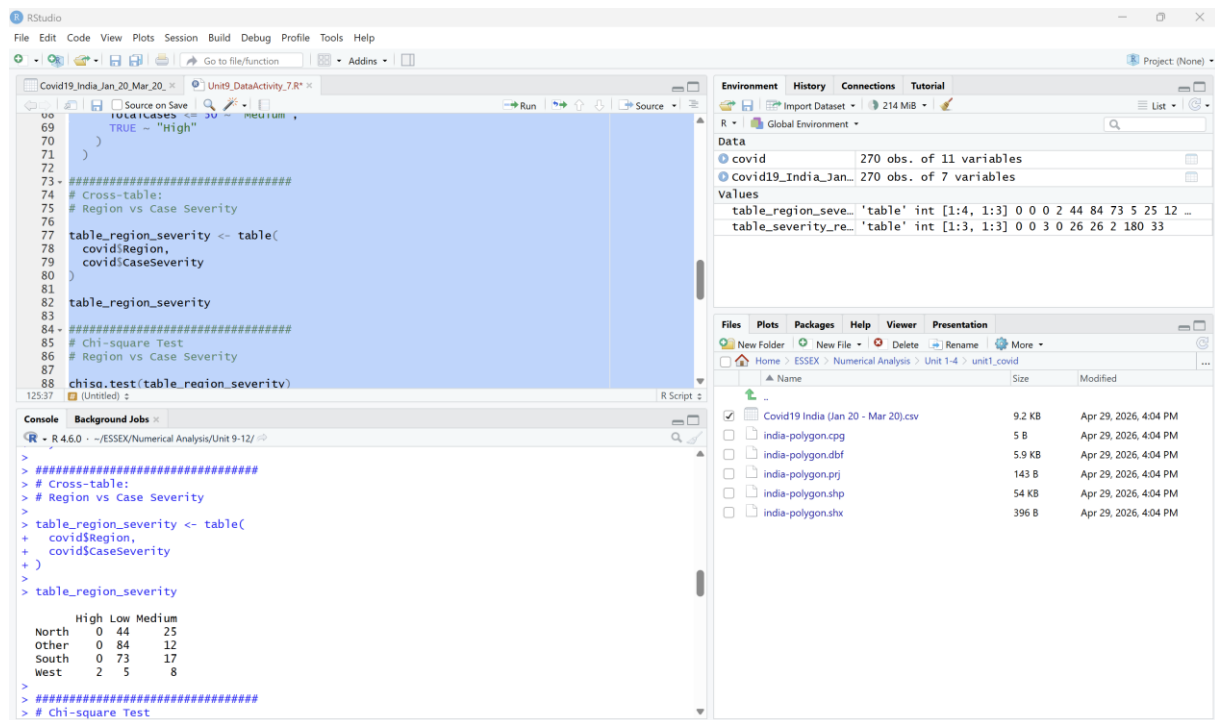
- medians are similar
- distributions overlap strongly
- no clear group separation is visible

SBP by Occupational Group



Kruskal-wallis rank sum test
data: sbp by occupation
Kruskal-wallis chi-squared = 0.77906, df = 3, p-value = 0.8545

Artefact 2 (Unit 9). Cross-tabulation and chi-square analysis in R: COVID-19 India reports grouped into Region and case-severity categories as set out in the activity brief (readr, dplyr). Evidence LO1 and LO2: implementing the specified groupings still required boundary decisions that shaped the resulting contingency tables.



Artefact 3 (Unit 10). Pearson correlation between systolic and diastolic blood pressure, Health_Data (n = 209): $r = 0.847$, $p < 2.2e-16$, 95% CI 0.804 to 0.881, with scatter plot. Evidence LO2 and LO3: focus on strength, direction, and uncertainty rather than making a binary decision.

Relationship Between Systolic and Diastolic Blood Pressure

1. CORRELATION ANALYSIS

Statistic	Result
Correlation coefficient (r)	0.847
Interpretation	Strong positive correlation
Meaning	As systolic blood pressure increases, diastolic blood pressure also tends to increase.
Sample size (n)	209
p-value (two-tailed)	< 2.2e-16
95% Confidence Interval for r	0.804 to 0.881
Conclusion	The correlation is statistically significant (p < 0.05). There is a strong positive relationship between systolic and diastolic blood pressure.

2. CORRELATION STRENGTH GUIDE (BY |r| VALUE)

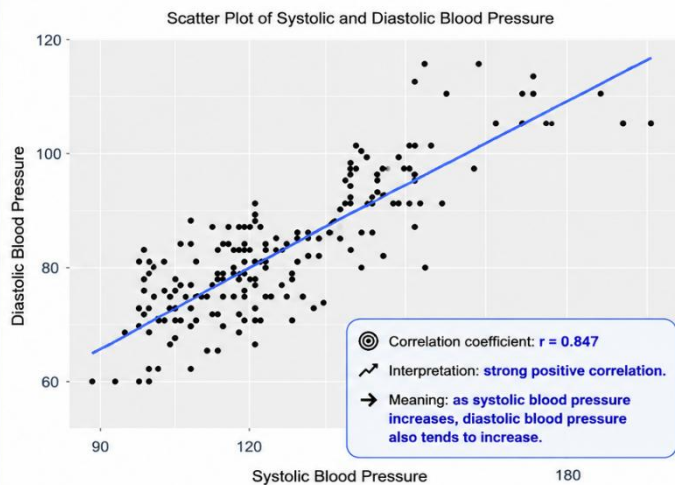
r Value Range	Strength of Relationship
0.00 – 0.19	Very weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very strong



KEY TAKEAWAY

Systolic and diastolic blood pressure are strongly and positively related. Monitoring one can help understand the other, which is important for early detection and management of cardiovascular risk.

3. SCATTER PLOT: SYSTOLIC vs DIASTOLIC BLOOD PRESSURE



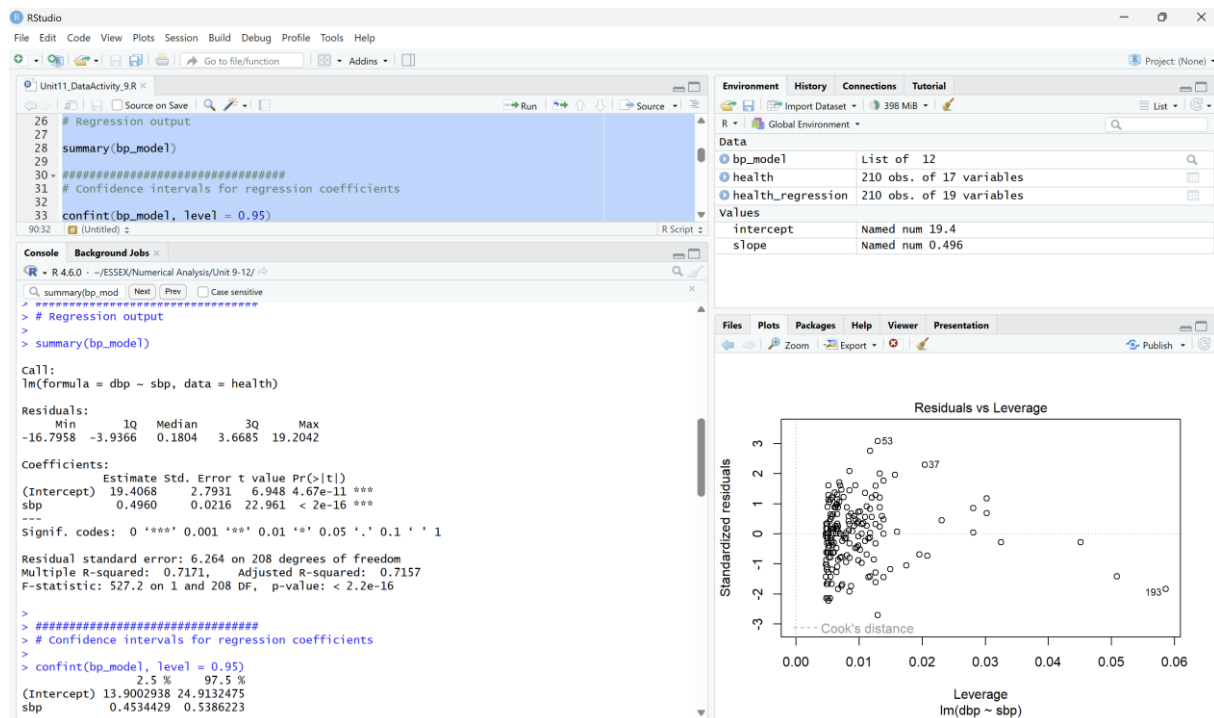
4. OVERALL CONCLUSION



The analysis shows a strong positive and statistically significant correlation between systolic and diastolic blood pressure ($r = 0.847$, $p < 2.2e-16$). This indicates that as systolic BP increases, diastolic BP also tends to increase. This relationship is important in clinical assessment and decision-making for cardiovascular health.

Method: Pearson correlation (two-tailed) | Software: R | Sample size (n): 209 | Confidence level: 95%

Artefact 4 (Unit 11). Simple linear regression predicting diastolic from systolic blood pressure: $DBP = 19.41 + 0.50 \times SBP$, $R^2 = 0.72$, slope 95% CI 0.45 to 0.54; residuals and leverage reviewed (observations 37, 53, 193). Evidence LO2 and LO3: model fit, coefficient interpretation and assumption checking.



Artefact 5 (Unit 12). Bayesian analysis of COVID-19 reporting: $P(\text{Kerala} \mid \text{Recovery})$ = 34.5% derived via Bayes' theorem from basic and conditional probabilities, computed in R. Evidence LO1 and LO3: updating belief about report origin as new evidence (recoveries) is observed.

COVID-19 INDIA REPORTS (JAN–MAR 2020): PROBABILITIES AND BAYES THEOREM

Understanding recovery reporting patterns in Kerala and Delhi

1. DATASET OVERVIEW



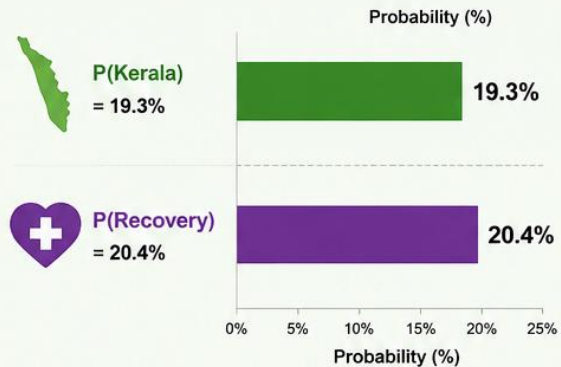
- Period: January – March 2020
- Source: COVID-19 India Dataset
- Total daily state reports: 270

Measure	Count
Kerala reports	52
Delhi reports	20
Reports with recoveries	55
Kerala reports with recoveries	19
Delhi reports with recoveries	8

These counts are validated from the dataset.

2. BASIC PROBABILITIES (Overall)

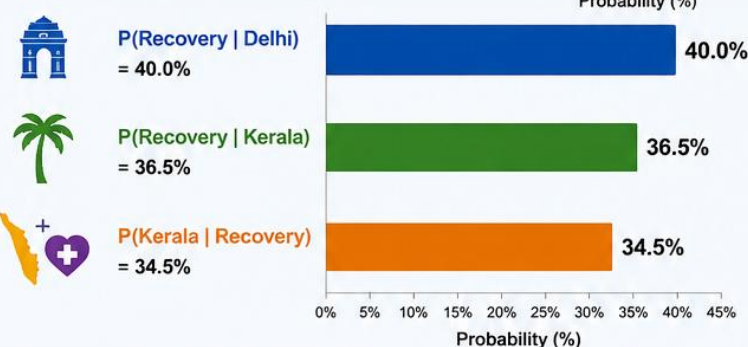
Probability of events without any conditions.



About 19.3% of all reports came from Kerala, and 20.4% of all reports showed patient recoveries.

3. CONDITIONAL PROBABILITIES & BAYES THEOREM

Probabilities of events given that another event has occurred.



BAYES THEOREM

The probability that a report came from Kerala given that it shows recoveries.

$$P(\text{Kerala} | \text{Recovery}) = \frac{P(\text{Recovery} | \text{Kerala}) \times P(\text{Kerala})}{P(\text{Recovery})}$$

STEP-BY-STEP CALCULATION

$$\begin{aligned}
 &= \frac{0.3654 \times 0.1926}{0.2037} \\
 &= 0.3455 \\
 &= \mathbf{34.55\%}
 \end{aligned}$$

4. WHAT DO THESE RESULTS MEAN?



Among Delhi reports, **40.0%** showed recoveries. Delhi had the highest recovery reporting rate.



Among Kerala reports, **36.5%** showed recoveries. Kerala also had a strong recovery reporting rate.



When a report showed recoveries, there was a **34.5%** chance that it came from Kerala. This is the result of Bayes theorem.



Overall, only **20.4%** of all reports contained recoveries. Recovery reporting was still developing in early 2020.

5. KEY TAKEAWAYS

- 1 Use basic probabilities to understand overall representation.
- 2 Use conditional probabilities to understand relationships between events.
- 3 Bayes theorem updates our belief about the origin of a report when new information (recoveries) is observed.
- 4 Kerala contributed a substantial share of the early recovery reports during the first phase of the COVID-19 pandemic in India.



This approach helps decision makers and researchers understand reporting patterns and make informed public health decisions.



Data: COVID-19 India Dataset (Jan–Mar 2020) | Total Reports = 270

Artefact 6 (Unit 5 formative discussion). Redesign of Brown (1994) Table 2 as a diverging stacked bar chart, with the peer critique received, my structured responses to two peers' charts, and a summary post developing the dot plot alternative raised in my initial post. Evidence LO4: appraising and presenting statistical results for a specific audience, and refining interpretation through exchange.

The screenshot shows a web browser window with the URL `my-course.co.uk/mod/forum/discuss.php?d=368957`. The browser's address bar and tabs are visible at the top. Below the browser window is the University of Essex online interface. On the left is a dark sidebar menu with the following items: Module Menu, Module Home, Tutor Office, Deadline Details, Seminars, Module Resources, e-Portfolio, eBook, Unit 1, Unit 2, Unit 3, and Unit 4. The main content area displays a forum post titled "Summary Post" by Anne Dayer, dated Monday, 15 June 2026, 6:38 PM. The post text discusses the redesign of Brown's (1994) Table 2, mentioning peer critiques and the author's structured responses. The text includes references to Kankaraš and Capecchi (2025), Feuerstahler's (2023) warning, and Panagiotis's caution. A "Chat to us!" button is visible at the bottom right of the post content. On the far right edge of the screenshot, there is a vertical purple bar with the text "Knowledge Base" and a small icon.

